Functional Testing: Designing Tests from the Specification

SE 413/513 Software Testing · Week 8

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Where we are

White-box (Weeks 5–7)

“Did my tests exercise the code I wrote?”

Control flow, data flow, coverage.py.

Functional (this week)

“Does the program do what the specification says?”

Decision tables, state machines, pairwise, properties.

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Four things coverage cannot see:

- Weak assertions
- Short-circuited sub-conditions
- Boundary values
- Missing features

What is functional testing?

Two meanings of the term

A test design approach

Tests derived from the *specified function (inputs, outputs, rules)*, without looking at the code. Same as black-box.

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Test oracle: the source of each expected result. In functional testing, *the specification*.

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The running specification

Checkout specification (excerpt)

REQ-1 Express shipping costs $10 for members and $15 for everyone else, regardless of order amount.
REQ-2 Standard shipping is free for members.
REQ-3 Standard shipping is free for any order of $50 or more.
REQ-4 Otherwise, standard shipping costs $7.99.
REQ-5 An order is created, then paid, then shipped, then delivered, in that order.
REQ-6 An order can be cancelled before it ships. Cancelling a paid order refunds the payment.
REQ-7 Any other action on an order is rejected and leaves the order unchanged.
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Week 7's checkout, extended with express shipping and an order lifecycle. Numbering is step one: every test can now name the requirement it checks.

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Conditions, cases, and traceability

Test condition: a testable aspect of the basis (“express-shipping for a member”).

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REQ-1–4	Decision table + BVA	`test_shipping_decision_table[R1..R5], test_free_shipping_boundary`
REQ-5–7	State transition testing	`test_valid_transitions, test_invalid_transitions_rejected, refund tests`
REQ-8	Property-based testing	`test_shares_add_up_to_total`

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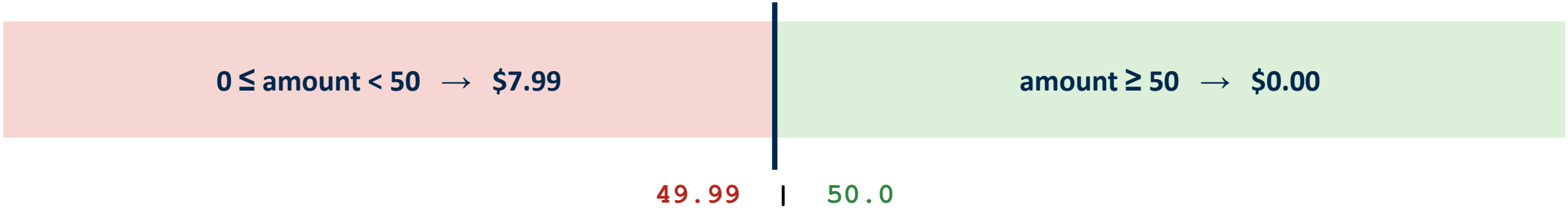
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Recap: partitions and boundaries

LIVE DEMO 1
boundary tests

REQ-3 and REQ-4, for a non-member choosing standard shipping:



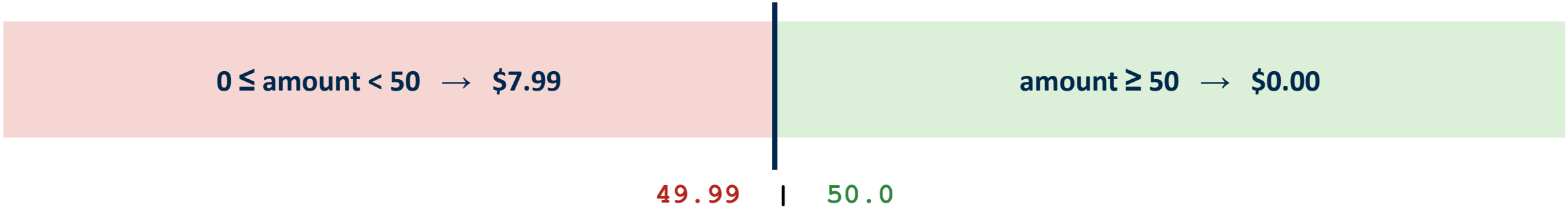
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REQ-3 and REQ-4, for a non-member choosing standard shipping:




```
@pytest.mark.parametrize("amount, expected",
                         [(49.99, 7.99), (50.0, 0.0)])
def test_free_shipping_boundary(amount, expected):
    assert shipping_fee(amount, False, False) == expected
```

EP and BVA work one input at a time. Next: combined rules, history, too many combinations, and "for all" requirements.

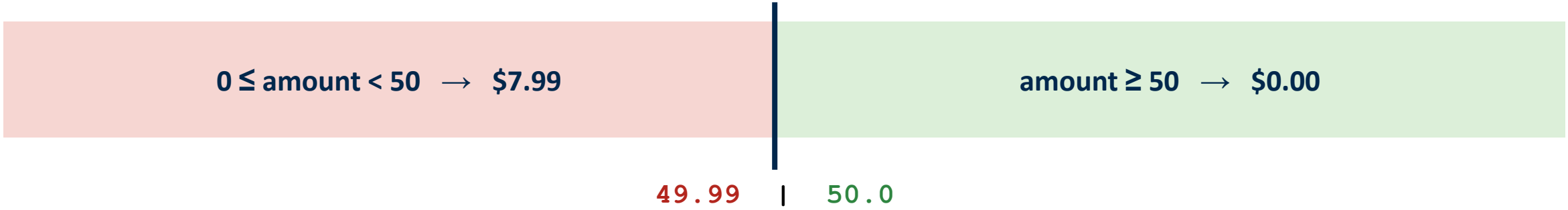
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*A **Decision Table** help us **grouping** Conditions/Rules*

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Decision tables: the full table

For rules shaped like “if this and that, but not the other, then X”. Three true/false conditions give $2^3 = 8$ rules (columns).

	1	2	3	4	5	6	7	8
C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
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3 Conditions ????

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For rules shaped like “if this and that, but not the other, then X”. Three true/false conditions give $2^3 = 8$ rules (columns).

	1	2	3	4	5	6	7	8
C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
C3: amount ≥ 50?	T	F	T	F	T	F	T	F
Fee $10.00 (REQ-1)	✓	✓						
Fee $15.00 (REQ-1)			✓	✓				
Fee $0.00 (REQ-2/3)					✓	✓	✓	
Fee $7.99 (REQ-4)								✓

Where spec bugs surface

Column 5 is free under both REQ-2 and REQ-3: consistent. Had REQ-2 said “$2.99 for members,” column 5 would expose a conflict before any code exists.

Collapse with “don't care” entries

Two columns that differ in one condition and share an action can merge: that condition gets “—”.

	1	2	3	4	5	6	7	8
C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
C3: amount ≥ 50?	T	F	T	F	T	F	T	F
Fee $10.00 (REQ-1)	✓	✓						
Fee $15.00 (REQ-1)			✓	✓				
Fee $0.00 (REQ-2/3)					✓	✓	✓	
Fee $7.99 (REQ-4)								✓

	R1	R2	R3	R4	R5
C1: express?	T	T	F	F	F
C2: member?	T	F	T	F	F
C3: amount ≥ 50?	—	—	—	T	F
Action: fee	**$10.00**	**$15.00**	**$0.00**	**$0.00**	**$7.99**
Full-table columns	1, 2	3, 4	5, 6	7	8

Decision table coverage = rules exercised by at least one test. One test per rule gives 100%.

Collapse with “don't care” entries

Two columns that differ in one condition and share an action can merge: that condition gets “—”.

	1	2	3	4	5	6	7	8
C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
C3: amount ≥ 50?	T	F	T	F	T	F	T	F
Fee $10.00 (REQ-1)	✓	✓						
Fee $15.00 (REQ-1)			✓	✓				
Fee $0.00 (REQ-2/3)					✓	✓	✓	
Fee $7.99 (REQ-4)								✓

	R1	R2	R3	R4	R5
C1: express?	T	T	F	F	F
C2: member?	T	F	T	F	F
C3: amount ≥ 50?	—	—	—	T	F
Action: fee	**$10.00**	**$15.00**	**$0.00**	**$0.00**	**$7.99**
Full-table columns	1, 2	3, 4	5, 6	7	8

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Two columns that differ in one condition and share an action can merge: that condition gets “—”.

	1	2	3	4	5	6	7	8
C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
C3: amount ≥ 50?	T	F	T	F	T	F	T	F
Fee $10.00 (REQ-1)	✓	✓						
Fee $15.00 (REQ-1)			✓	✓				
Fee $0.00 (REQ-2/3)					✓	✓	✓	
Fee $7.99 (REQ-4)								✓

	R1	R2	R3	R4	R5
C1: express?	T	T	F	F	F
C2: member?	T	F	T	F	F
C3: amount ≥ 50?	—	—	—	T	F
Action: fee	**$10.00**	**$15.00**	**$0.00**	**$0.00**	**$7.99**
Full-table columns	1, 2	3, 4	5, 6	7	8

Decision table coverage = rules exercised by at least one test. One test per rule gives 100%.

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C1: express?	T	T	T	T	F	F	F	F
C2: member?	T	T	F	F	T	T	F	F
C3: amount ≥ 50?	T	F	T	F	T	F	T	F
Fee $10.00 (REQ-1)	✓	✓						
Fee $15.00 (REQ-1)			✓	✓				
Fee $0.00 (REQ-2/3)					✓	✓	✓	
Fee $7.99 (REQ-4)								✓

	R1	R2	R3	R4	R5
C1: express?	T	T	F	F	F
C2: member?	T	F	T	F	F
C3: amount ≥ 50?	—	—	—	T	F
Action: fee	**$10.00**	**$15.00**	**$0.00**	**$0.00**	**$7.99**
Full-table columns	1, 2	3, 4	5, 6	7	8

Decision table coverage = rules exercised by at least one test. One test per rule gives 100%.

Collapse with “don't care” entries

Two columns that differ in one condition and share an action can merge: that condition gets “—”.

	R1	R2	R3	R4	R5
C1: express?	T	T	F	F	F
C2: member?	T	F	T	F	F
C3: amount ≥ 50?	—	—	—	T	F
Action: fee	**$10.00**	**$15.00**	**$0.00**	**$0.00**	**$7.99**
Full-table columns	1, 2	3, 4	5, 6	7	8

8 → 5

rules after collapsing

Careful

“—” is the spec's claim that a condition is irrelevant. The code may disagree.

Decision table coverage = rules exercised by at least one test. One test per rule gives 100%.

One test per rule

LIVE DEMO 2
run the rules

```
@pytest.mark.parametrize("amount, is_member, express, expected", [
    pytest.param(20.0, True,  True,  10.0, id="R1-member-express"),
    pytest.param(20.0, False, True,  15.0, id="R2-guest-express"),
    pytest.param(20.0, True,  False, 0.0,  id="R3-member-standard"),
    pytest.param(80.0, False, False, 0.0,  id="R4-guest-standard-50plus"),
    pytest.param(20.0, False, False, 7.99, id="R5-guest-standard-under50"),
])
def test_shipping_decision_table(amount, is_member, express, expected):
    assert shipping_fee(amount, is_member, express) == expected
```

```
$ pytest tests/test_shipping_dt.py -v
...test_shipping_decision_table[R1-member-express] PASSED
...test_shipping_decision_table[R2-guest-express] PASSED
...test_shipping_decision_table[R3-member-standard] PASSED
...test_shipping_decision_table[R4-guest-standard-50plus] PASSED
...test_shipping_decision_table[R5-guest-standard-under50] PASSED
```

Rule IDs show up in the report, so a failure names the rule it broke.

What it catches: rule priority

LIVE DEMO 3
break the priority

shop/shipping_buggy.py: free shipping checked first

```
def shipping_fee(amount, is_member, express):
    if is_member or amount >= 50:
        return 0.0
    if express:
        return 10.0 if is_member else 15.0
    return 7.99
```

```
FAILED ...[R1-member-express]
1 failed, 6 passed
```

R1 fails: a member's express order is free instead of $10.

“Don't care” is not “don't test”

The same bug makes a guest's $80 express order free. R2 was tested only at $20, where the bug does not show. When cheap, test the — condition both ways.

Table + BVA

The table picks combinations; BVA picks values. Test R4 and R5 at 50.0 and 49.99.

What it catches: rule priority

LIVE DEMO 3
break the priority

shop/shipping_buggy.py: free shipping checked first

```
def shipping_fee(amount, is_member, express):
    if is_member or amount >= 50:
        return 0.0
    if express:
        return 10.0 if is_member else 15.0
    return 7.99
```

```
FAILED ...[R1-member-express]
1 failed, 6 passed
```

R1 fails: a member's express order is free instead of $10.

“Don't care” is not “don't test”

The same bug makes a guest's $80 express order free. R2 was tested only at $20, where the bug does not show. When cheap, test the — condition both ways.

Table + BVA

The table picks combinations; BVA picks values. Test R4 and R5 at 50.0 and 49.99.

Try it: don't care, but test it

TRY IT 1
students · 4 min

1. Point the decision-table tests at the buggy version. Which rule fails?

```
# tests/test_shipping_dt.py:
from shop.shipping_buggy import shipping_fee
pytest tests/test_shipping_dt.py
```

2. Add a case that tests R2's don't-care condition at $80. Predict the result first.

```
pytest.param(80.0, False, True, 15.0,
             id="R2-guest-express-80")
```

3. Run again on the buggy version, then on the correct one. Switch the import back.

```
pytest tests/test_shipping_dt.py
# then: from shop.shipping import shipping_fee
```

Try it: build a decision table

TRY IT 2
students · 5 min

1. A student may register if prerequisites are met AND the section has seats AND there is no account hold.

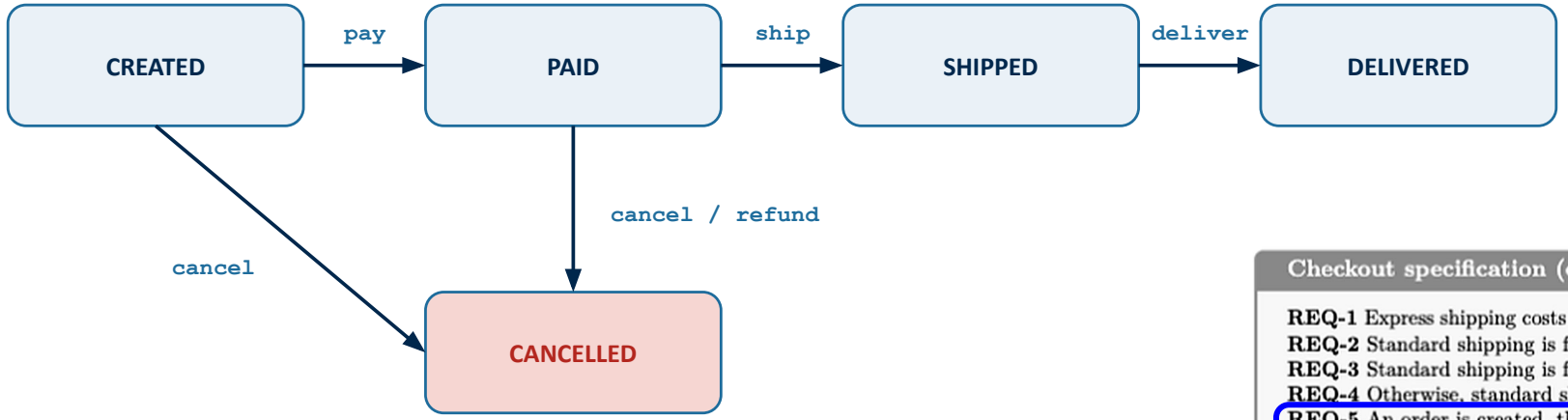
2. If prerequisites are met and there is no hold but the section is full, the student is waitlisted. Otherwise: rejected.

3. On paper: (a) build the full table, (b) collapse it. How many rules are left?

Done early? Write the parametrized pytest test with one rule ID per case (Exercise 1c in the reading).

State transition testing: the model

When the right answer depends on history: carts, sessions, approvals, wizards. REQ-5–7 describe an order lifecycle.



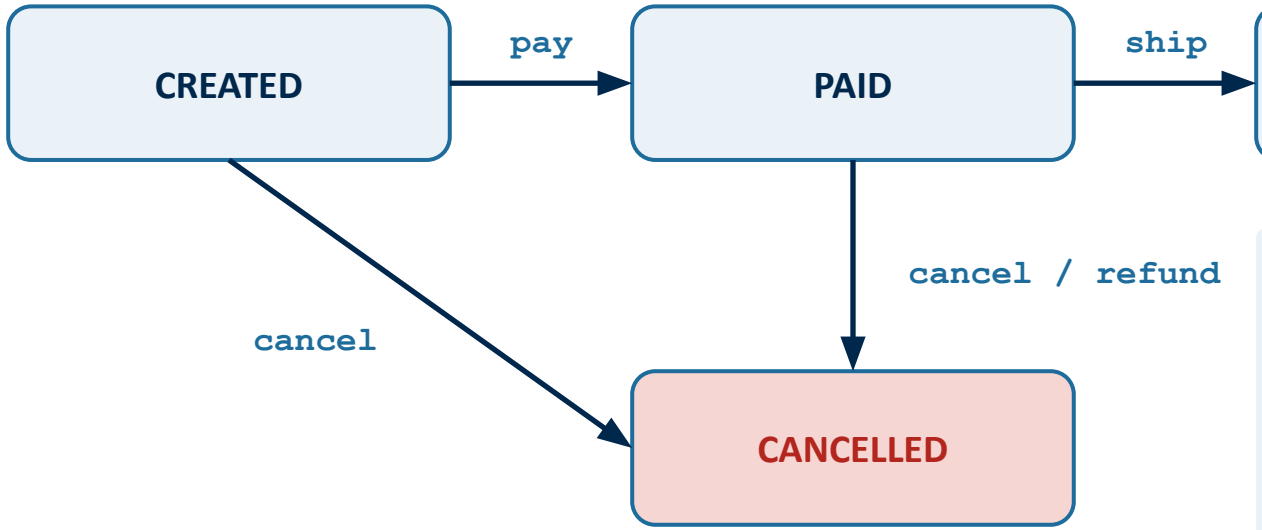
Checkout specification (excerpt)

REQ-1 Express shipping costs $10 for members and $15 for everyone else, regardless of order amount.
REQ-2 Standard shipping is free for members.
REQ-3 Standard shipping is free for any order of $50 or more.
REQ-4 Otherwise, standard shipping costs $7.99.
REQ-5 An order is created, then paid, then shipped, then delivered, in that order.
REQ-6 An order can be cancelled before it ships. Cancelling a paid order refunds the payment.
REQ-7 Any other action on an order is rejected and leaves the order unchanged.
REQ-8 A bill split among n people produces n shares, in cents, that add up exactly to the total.

5 states, 4 events, 5 valid transitions. Anything not in the dictionary raises InvalidTransition.

State transition testing: the model

When the right answer depends on history: carts, sessions, approvals, wizards. REQ-5–7 describe an order lifecycle.




```
_TRANSITIONS = {
    ("CREATED", "pay"): "PAID",
    ("CREATED", "cancel"): "CANCELLED",
    ("PAID", "ship"): "SHIPPED",
    ("PAID", "cancel"): "CANCELLED",
    ("SHIPPED", "deliver"): "DELIVERED",
}
```

5 states, 4 events, 5 valid transitions. Anything not in the dictionary raises InvalidTransition.

The state table: finding invalid transitions

State \ Event	pay	ship	deliver	cancel
CREATED	**PAID**	invalid	invalid	**CANCELLED**
PAID	invalid	**SHIPPED**	invalid	**CANCELLED + refund**
SHIPPED	invalid	invalid	**DELIVERED**	invalid
DELIVERED	invalid	invalid	invalid	invalid
CANCELLED	invalid	invalid	invalid	invalid

20

(state, event) pairs

15

invalid: REQ-7 says reject

The diagram shows only what should happen. Paying twice, shipping a cancelled order, cancelling after delivery: the defects hide in the red cells.

Coverage criteria and a test helper

Criterion	Tests must…	Order
All states	visit every state	5 states
Valid transitions (0-switch)	fire every valid transition	5 transitions
All transitions	also try every invalid one and check it is rejected	**20 pairs**

Each is stronger than the one above it. 1-switch would test every pair of consecutive transitions.

Reach a state only through the public API

```
def order_in(state):
    o = Order()
    path = {"CREATED": [],
            "PAID": ["pay"],
            "SHIPPED": ["pay", "ship"],
            "DELIVERED": ["pay", "ship",
                          "deliver"],
            "CANCELLED": ["cancel"]}[state]
    for event in path:
        getattr(o, event)()
    return o
```

Never o.state = "SHIPPED": it skips the very transitions under test.

All-transitions tests

LIVE DEMO 4
22 state tests

```
@pytest.mark.parametrize("start, event, end", VALID_CASES)    # 5 cases
def test_valid_transitions(start, event, end):
    o = order_in(start)
    getattr(o, event)()
    assert o.state == end

@pytest.mark.parametrize("start, event", INVALID)             # 15 pairs
def test_invalid_transitions_rejected(start, event):
    o = order_in(start)
    with pytest.raises(InvalidTransition):
        getattr(o, event)()
    assert o.state == start          # REQ-7: order unchanged
```

```
$ pytest tests/test_order.py
22 passed
```

22

5 valid + 15 invalid + 2 refund tests

Check both halves

Rejecting the event is not enough. REQ-7 also says the order is unchanged, so the test asserts the state too.

Try it: add a return

TRY IT 3
students · 5 min

1. New event return_item: valid only from DELIVERED, moves to a new RETURNED state, refunds the payment.

2. Update the diagram and state table. How many (state, event) pairs now? How many invalid?

```
states: 5 + RETURNED
events: 4 + return_item
pairs = states x events = ?
```

3. Which tests change to keep all-transitions coverage?

Done early? Implement it in shop/order.py and extend tests/test_order.py. How many tests pass?

The combinatorial explosion

Parameter	Values
Payment method	`card, paypal, gift_card`
Delivery	`standard, express, pickup`
Customer	`member, guest`
Coupon	`none, SAVE10, FLAT5`

54

3 × 3 × 2 × 3 end-to-end tests

216

with one more 4-valued option

Exhaustive testing grows multiplicatively, and quickly becomes impractical for anything slower than a unit test.

Pairwise (2-way) testing

Every pair of values, for every pair of parameters, appears together in at least one test. Each test covers many pairs: 4 parameters give 6 pairs per test.

Generating a pairwise set with allpairspy

LIVE DEMO 5
generate + verify

```
from allpairspy import AllPairs

params = [
    ["card", "paypal", "gift_card"],
    ["standard", "express", "pickup"],
    ["member", "guest"],
    ["none", "SAVE10", "FLAT5"],
]

@pytest.mark.parametrize(
    "payment, delivery, customer, coupon",
    [tuple(r) for r in AllPairs(params)])
def test_checkout_pairwise(...): ...
```

#	Payment	Delivery	Customer	Coupon
1	card	standard	member	none
2	paypal	express	guest	none
3	gift_card	pickup	guest	SAVE10
4	gift_card	express	member	FLAT5
5	paypal	standard	member	SAVE10
6	card	pickup	guest	FLAT5
7	card	express	guest	SAVE10
8	paypal	pickup	member	FLAT5
9	gift_card	standard	guest	FLAT5
10	gift_card	pickup	guest	none

10 tests instead of 54; a check script confirms all 45 pairs appear. Minimum possible: 9.

Why pairwise works, and where it stops

The NIST interaction rule

Kuhn, Wallace & Gallo (IEEE TSE, 2004) studied failure reports from real systems, including medical-device software and a web browser.

Most failures needed one parameter value or an interaction of two. Fewer needed three or more. None needed more than six.

! **3+ way failures:** Not guaranteed to be found. For critical features, use t-way (e.g., NIST ACTS).

! **No values:** EP and BVA still choose the values for each parameter.

! **Independence:** Invalid combinations must be excluded (allpairspy has a filter function).

! **No oracle:** Each row still needs an expected result.

Try it: add a parameter

TRY IT 4
students · 4 min

1. Add currency (USD, CAD, EUR, GBP) as a fifth parameter. How many exhaustive combinations?

```
    ["USD", "CAD", "EUR", "GBP"],    # 5th list
```

2. Generate the pairwise set with allpairspy and report its size.

```
rows = list(AllPairs(params))
print(len(rows))
```

3. Exclude gift_card with a non-USD currency using a filter function.

```
def ok(row):
    return not (len(row) > 4 and row[0] == "gift_card"
                and row[4] != "USD")
AllPairs(params, filter_func=ok)
```

Done early? Extend pairwise.py's check to the filtered set: is every valid pair still covered?

From examples to properties

REQ-8 is about all inputs: for any total and any n, the shares add up exactly to the total.

shop/split.py: first version

```
def split_bill(total, people):
    share = round(total / people, 2)
    return [share] * people
```

Example tests pass: split_bill(30.0, 3) gives three $10 shares; split_bill(100.0, 4) gives four $25 shares.

@given calls the test many times (100 by default) with inputs drawn from the strategies.

tests/test_split_pbt.py

```
from hypothesis import given, strategies as st
from shop.split import split_bill

@given(
    total_cents=st.integers(min_value=0,
                            max_value=100_000),
    people=st.integers(min_value=1, max_value=10),
)
def test_shares_add_up_to_total(total_cents, people):
    total = total_cents / 100
    shares = split_bill(total, people)
    assert len(shares) == people
    assert round(sum(shares), 2) == total
```

Hypothesis finds a counterexample

LIVE DEMO 6
find the bug

```
$ pytest tests/test_split_pbt.py
E       assert 0.02 == 0.01
E        +  where 0.02 = round(0.02, 2)
E        +    where 0.02 = sum([0.01, 0.01])
E       Failing test case: test_shares_add_up_to_total(
E           total_cents=1,
E           people=2,
E       )
FAILED tests/test_split_pbt.py::test_shares_add_up_to_total
```

1¢ ÷ 2

two one-cent shares: the bill grows

Any total that does not divide evenly fails (try $100 among 3).

Shrinking: Hypothesis reduces a failing input to the simplest one that still fails, so it is easy to understand.

Memory: Failing examples are saved in .hypothesis/ and retried first on the next run.

The fix, and finding good properties

LIVE DEMO 7
fix + re-run

shop/split_fixed.py: work in integer cents

```
def split_bill(total, people):
    cents = round(total * 100)
    base, extra = divmod(cents, people)
    return [(base + 1) / 100 if i < extra
            else base / 100
            for i in range(people)]
```

```
>>> split_bill(0.01, 2)
[0.01, 0.0]
>>> split_bill(100, 3)
[33.34, 33.33, 33.33]
```

The property test now passes (1 passed).

Pattern	Example
Invariant	shares add up to the total
Round trip	`decode(encode(x)) == x`
Idempotence	normalizing twice = normalizing once
Model	fast version agrees with a slow, obvious one
Metamorphic	adding an item never lowers a subtotal

Pitfall: a property that re-implements the function (== [round(t/n, 2)] * n) copies its bug and can never fail. State what must be true of the result.

Try it: properties for apply_coupon

TRY IT 5
students · 5 min

1. Write two Hypothesis properties for Week 7's apply_coupon(amount, code), for non-negative amounts.

```
amounts = st.floats(min_value=0, max_value=10_000)
codes = st.sampled_from(["SAVE10", "FLAT5", "NONE"])

@given(amounts, codes)
def test_...(amount, code): ...
```

2. Suggested: the result is never negative; a coupon never raises the price.

```
assert apply_coupon(amount, code) >= 0
assert apply_coupon(amount, code) <= amount
```

3. Run them. Does either find a counterexample? Why?

```
pytest test_coupon_props.py
```

Choosing a technique

If the specification…	Use	Python support
has ranges or categories of one input	EP + BVA (Week 3)	`parametrize`
combines several conditions into rules	Decision table	`parametrize + rule IDs`
depends on history or mode	State transition	`parametrize over (state, event)`
has many independent options	Pairwise / t-way	`allpairspy`
says “always” or “for any”	Property-based	`Hypothesis`

Functional testing + coverage: run the spec-based suite under pytest-cov. Code it never executes is either behavior the spec does not mention, or a requirement your tests missed.

Common mistakes

! The code as the oracle

Expected values come from the spec, not from running the code.

! Untested don't-cares

The code may not agree a condition is irrelevant.

! Happy-path state tests

Most of the 20 pairs are invalid, and that is where defects hide.

! Setting state directly

o.state = … bypasses the transitions under test.

! Pairwise as exhaustive

It covers 2-way interactions, not all failures.

! Properties that restate code

Describe the result, not the algorithm.

A practical mental model

When you see: **“if … and … unless …”**	Think: Decision table. How many conditions? Is every column filled in?
When you see: **The answer depends on what happened before**	Think: State machine. What should happen in every cell of the state table?
When you see: **The combinations make you wince**	Think: Pairwise first; higher strength where failure is expensive.
When you see: **“always” or “for any”**	Think: A property. Let Hypothesis look for a counterexample.

Key takeaways

1. The specification is the oracle, never the current behavior of the code.
2. Number requirements and trace tests to them.
3. Decision tables for combined conditions; test the don't-cares.
4. State tables expose the invalid transitions where defects hide.
5. Pairwise tames the explosion; most failures involve one or two parameters.
6. Properties check “for all” requirements; Hypothesis shrinks failures.
7. Combine techniques, and combine them with coverage.

BK

Learning objectives

1 Oracle

Explain functional testing and why the specification is the oracle.

2 Trace

Turn a specification into numbered requirements and traceable tests.

3 Tables

Build and collapse a decision table, then derive one test per rule.

4 States

Model a state machine and test valid and invalid transitions.

5 Pairs

Reduce combinations with pairwise testing (allpairspy).

6 Properties

Test “for all” requirements as properties with Hypothesis.